

MERN Stack Development – Module 4 Exercise

Hands-on Exercise No. 4

Course: MERN Stack Development

Module: Deployment

Total Marks: 10

Due Date: 20/05/2026

Instructions:

- Prepare your solution in MS Word
 - Submit in .doc or .docx format
 - Late or copied submissions will not be accepted
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Learning Outcome:

- Deploy MERN applications on cloud platforms
 - Configure environment variables securely
 - Implement CI/CD pipelines
 - Understand application scalability
 - Compare different hosting platforms
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Problem Statement:

You are working as a Full Stack Developer for a startup planning to launch a MERN-based eCommerce application. The client requires the application to be deployed online with automatic updates and scalable infrastructure.

Task Requirements:

- Deploy a sample MERN application
 - Backend on Heroku
 - Frontend on Netlify
- Configure environment variables securely
- Set up a CI/CD pipeline using GitHub Actions or Jenkins
- Demonstrate automatic deployment when code is pushed to the repository
- Test the live application using the deployed URL

- Take screenshots of:
 - Heroku dashboard
 - Netlify dashboard
 - CI/CD pipeline execution
 - Live application running in browser
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Submission Guidelines:

- Upload your .docx file on LMS
- Include:
 - Deployment steps
 - Screenshots of all required sections
 - CI/CD configuration details
- Add short explanations for each step